POSTER

# Poster: Design of Mixed Reality Dangerous Situations for Autistic Children: Road Safety

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# Poster: Design of Mixed Reality Dangerous Situations for Autistic Children: Road Safety

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**ABSTRACT**

While many road safety programs have been instituted over the past decades, bringing down the rate of injury for children from 3.6 deaths per 100,000 to .3, still, every year in the United States, approximately 600 child pedestrians are killed and 76,000 child pedestrians are injured. Children on the autistic spectrum often require additional considerations than neuro-typical children when trying to teach road safety skills, including differences in thinking flexibility, difficulty in understanding social contexts and communication, and over-sensitivity or under-sensitivity to sights and sounds. A mixed reality road safety environment design is proposed to address several considerations, with potential technical and clinical design challenges.

## CCS CONCEPTS

• Information Systems • Information Systems Applications

## KEYWORDS

Mixed Reality (MR), Augmented Reality (AR), human-machine interface, Autism Spectrum Disorder (ASD), education

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## 1 Introduction and Motivation

Developing learning curricula and material for dangerous situations is challenging for many scenarios and people, from toddlers to adults. Children with Autism Spectrum Disorder (ASD), which is a social communication disorder, have increased support requirements that make learning situations more difficult and are three times more likely to die from injuries than the general population [1]. While studies show the promising educational value of utilizing augmented reality (AR) for children with ASD, little to no research has been conducted on using mixed reality (MR) for ASD educational purposes, and neither technology has been researched significantly for the specific purpose of teaching children with ASD how to handle dangerous situations [2].

Mixed reality has emerged as interesting technology that allows for the presentation of 3-dimensional (3D) holograms that can be "hooked" to features of users' real-world environment, enabling presence and interaction similar to interacting as if the 3D elements were physically present, including pushing, pressing, and spatially accurate sounds. This includes having multiple users in the same local environment experiencing the 3D objects as having relative position to the environment and not to just themselves, i.e. if a 3D elephant is created and standing in a location in a room for one user, the 3D elephant will be spatially presented with the appropriate viewpoint for other users that are in a different physical location from one another. This creates opportunities for *simulation within a real-world environment* that has not previously been possible. This includes simulating dangerous situations with 3D elements that can potentially provide visual, audio, and tactile feedback that would provide training cues for such situations, without placing the users in real danger.

These features of MR provide potential benefits for developing MR training scenarios for children with ASD, including the following reasons. First, children with ASD may have difficulties with behaving flexibly and may not generalize new skills to various situations; MR training scenarios have the potential to be quickly adapted to integrate different 3D elements and to place those elements in different relative spatial orientation quickly, thus enabling reinforcement of learned skills that may allow for better opportunities to generalize. Second, children with ASD may become easily distracted, which minimizes potential learning situations; MR would allow for providers to reset, rewind, pause, or make other time related modifications, enabling engagement at a desired time point and location that would provide opportunities to children who require multiple attempts to learn skills with multiple steps. Third, children with ASD may have over-sensitivity, or under-sensitivity, to sights and sounds; MR presentation of 3D elements can be programmatically adapted for various attributes that may allow for better sensitization in multiple conditions, including varying speed, size, and sound of the 3D elements, enabling the potential to "work up" a child's ability to handle situations. Fourth, children with ASD often have difficulty understanding context which makes understanding right and wrong difficult; MR would allow for interaction in a realistic manner with both 3D elements and real-world elements in a way


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that would allow learning opportunities on how to act in different social situations that create explicit instruction of "teachable moments" in realistic settings.

Pedestrian road safety is vital for all individuals, with approximately 76,000 child pedestrians suffering injuries annually and approximately 600 receiving fatal injuries. Traditional road safety programs are conducted at all levels to increase road safety awareness. However, programs that can accommodate all of the previously mentioned challenges that children with ASD may have, are not available without extreme logistical and technical challenges, with aspects that may be impossible to address. As MR systems possess the potential to address several challenging areas for children with ASD, a mixed-reality road safety environment is proposed.

## 2 Design of MR Road Safety Environment

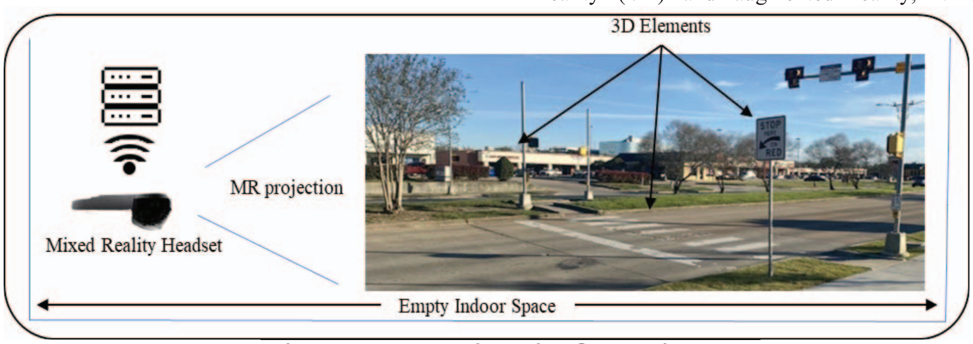


Figure 1. MR-Based Road Safety Environment

Fig. 1 shows the design of an MR road safety environment, consisting of two major components, an MR headset and server. Additionally, the depicted scenario is such that the physical portion of the MR environment is an empty indoor space (e.g. a gymnasium). The headset delivers the visual and audio aspects features of various 3D elements, which in Fig. 1 contains a roadway, sidewalk, crosswalk, signage, and walk signals. The MR is "hooked" to the indoor space, such that any user immersed in the MR environment is spatially placed within that environment relative to the depicted environment (i.e. the environment is stable and able to be traversed). The server is used to store data collected about the users' actions for further data analysis (e.g., eye-tracking, response times, etc.). 3D elements are local to the storage system contained within the MR headsets (HoloLens 2). Typical usage would be two MR headsets. One for the client (patient), the other for the therapist. More MR headsets could be utilized as required. The MR environment is developed within the Unity framework. A "floating" menu-based user interface (UI) allows for 3D element orientation, position, and timing control. Vehicles within the orientation are "smart" elements that react to the traffic control signals and any other vehicles within the environment.

## 3 Technical and Clinical Design Challenges

Control of the UI for Unity based MR is typically through "floating" menus or voice commands. As the desired UI is menu-driven to avoid unintended UI activations, and current Unity MR does not allow for 3D elements to only be visible for a single MR headset, the control of UI elements is restricted to the MR headset designated for the therapist, with the ability to hide the UI as-needed to maintain the MR road safety environment realism.

For clinical use, potential considerations for children with ASD training in the road safety environment include cases where the child may learn potential behavior that might lead to real injury. As the 3D elements do not currently have tactile feedback, it is possible that a child might "learn" that walking into traffic has no physical consequence. Therefore, a design requirement for collision detection between an MR headset and a moving vehicle that results in three user-selectable options: the MR environment freezes, the MR environment disappears, or the MR environment indicates that a hazard condition has occurred by flashing the environment in red accompanied by an alarm sound. The choice of action is based upon the therapists' desired training outcome for the individual.

With very little in the literature regarding MR effectiveness for children with ASD, but some preliminary information with virtual reality (VR) and augmented reality, it is anticipated that participants may find using MR technology difficult in various ways. This includes feeling uncomfortable wearing the MR headset, distraction of MR elements, and ease of use of the MR UI [2]. However, MR does provide potential to avoid a reported issue with VR training being less effective when the VR environment tries to emulate a known environment to participants, when the two environments do not "match". As MR can utilize a "real" location while including MR 3D elements, it is possible that this reported distraction effect can be reduced.

## 4 Conclusion

MR has great potential to allow safe and controlled training for multiple scenarios, including dangerous scenarios. A road safety environment scenario design has been presented, with a focus on the needs of children with ASD for use in a clinical setting. An initial version of the system is expected to be pilot tested to determine usability and acceptance, for clinicians and patients. It is expected that the feedback received will allow for continued refinement that will allow for a larger-scale test of its effectiveness within different populations, including children with ASD. If proven to be effective, this MR road safety environment would allow a realistic way for training that would not usually be expected for children with ASD, and develop valuable lessons for designing other MR dangerous scenarios.

## REFERENCES

[1] Joseph Guan and Guohua Li. 2017. Injury Mortality in Individuals with Autism. *American Journal of Public Health.* 107, 5 (May 2017) 791-793. DOI: https://doi.org/10.2105/AJPH.2017.303696

[2] Maggie A. Mosher and Adam C. Carreon. 2021. Teaching social skills to students with autism spectrum disorder through augmented, virtual, and mixed reality. *Research in Learning Technology.* 29 (August 2021). DOI:http://dx.doi.org/10.25304/rlt.v29.2626